

THE QUANTUM ALGORITHMS ROADMAP

A COMPREHENSIVE SEQUENCE FROM FOUNDATIONS TO RESEARCH



PHASE 1

Foundational Oracle Algorithms

PROOFS OF CONCEPT



Deutsch-Jozsa

The first algorithm to prove quantum advantage.

Distinguishes constant from balanced functions in a single query.



Bernstein-Vazirani

Retrieves a hidden bitstring in $O(1)$ complexity compared to classical $O(n)$, proving oracle speedups.



Simon's Algorithm

Identifies periodicity in functions. This serves as the critical mathematical precursor to Shor's algorithm.

PHASE 2

Exponential Speedups

ALGEBRAIC & PERIODIC LOGIC

Quantum Fourier Transform

The core engine for algebraic speedups. Maps quantum amplitudes to the frequency domain exponentially faster than classical FFT.

Shor's Algorithm

Utilizes QFT for period finding to factor large integers in $O(n^3)$ time, directly challenging RSA-based classical cryptography.

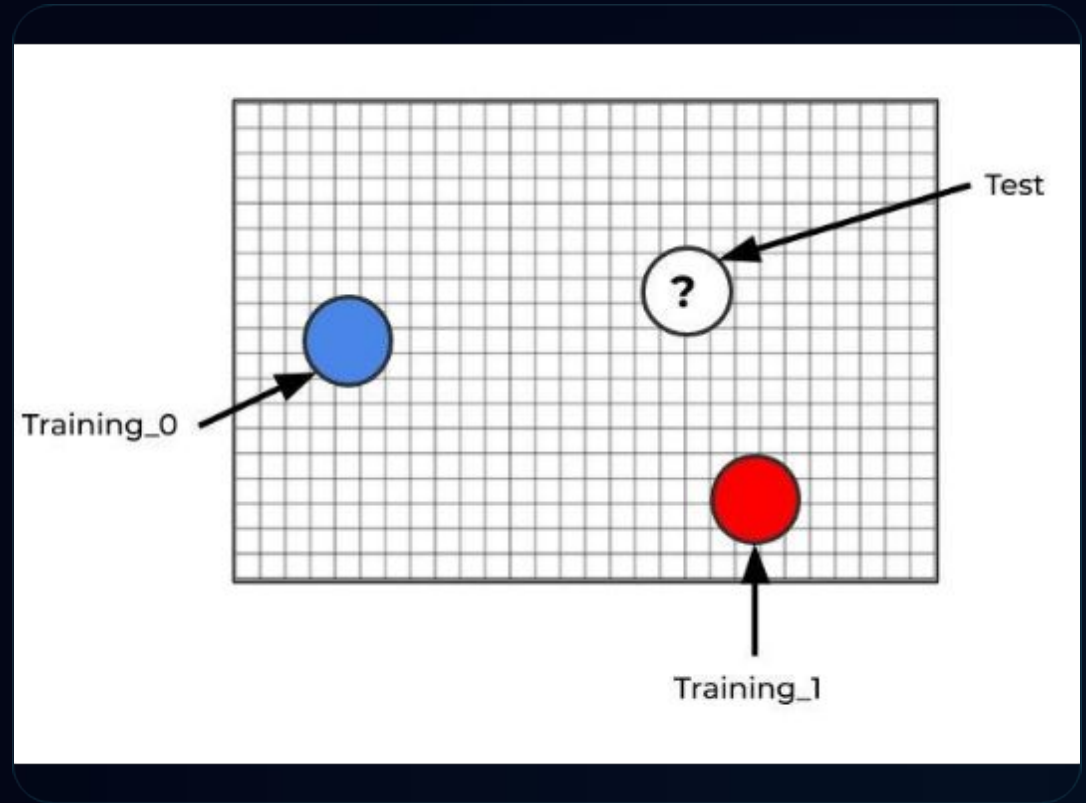
SEARCH & COMPLEXITY

Grover's Algorithm

Achieves a quadratic speedup for unstructured searching.

Where classical search requires $O(N)$ operations, Grover requires only $O(\sqrt{N})$.

This provides a broad advantage for NP-hard problems and database queries across diverse domains.



PHASE 3

The NISQ
Era

VARIATIONAL QUANTUM ALGORITHMS



VQE

Variational Quantum Eigensolver. Primary tool for quantum chemistry and molecular energy simulation on noisy hardware.



QAOA

Quantum Approximate Optimization Algorithm. Solves combinatorial problems like Max-Cut through hybrid training.



Hybrid Loops

Leveraging classical optimizers to update quantum circuits, mitigating noise effects in intermediate systems.



PHASE 4

The Future Frontier

LINEAR SYSTEMS & BIG DATA



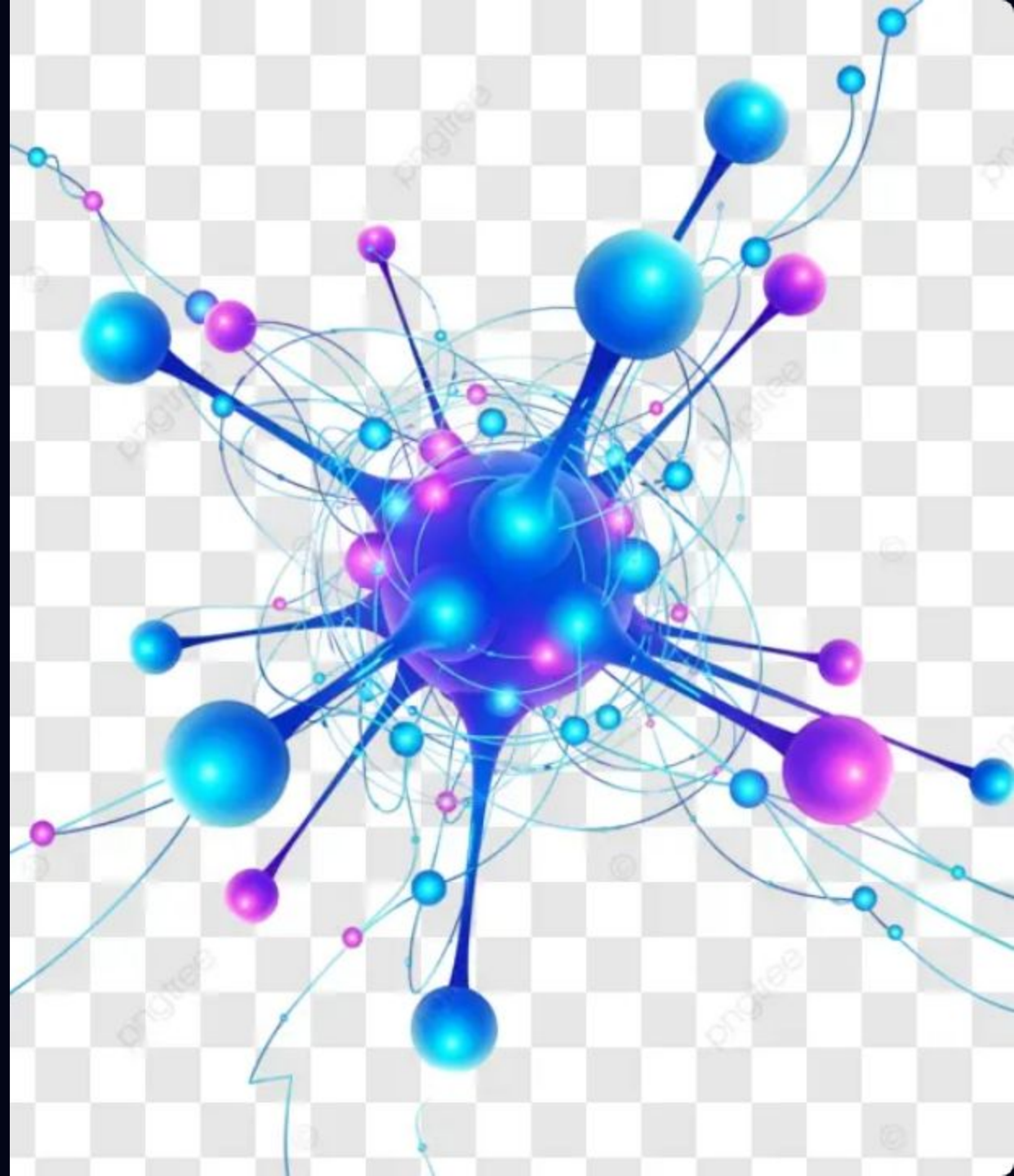
Comparison of computational complexity for solving massive linear systems.

QUANTUM ML

Advanced QML

Utilizing Quantum Kernels and Feature Maps to process high-dimensional datasets that are classically intractable.

Recent research focuses on "Barren Plateaus" mitigation and data-efficient training of Quantum Neural Networks (QNNs).



DEVELOPMENT ROADMAP

